

DEVELOPMENT OF A CRITERION METHOD TO DETERMINE PEAK MECHANICAL POWER OUTPUT IN A COUNTERMOVEMENT JUMP

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ABSTRACT

Owen, NJ, Watkins, J, Kilduff, LP, Bevan, HR, and Bennett, MA. Development of a criterion method to determine peak mechanical power output in a countermovement jump. *J Strength Cond Res* 28(6): 1552–1558, 2014—There is a general agreement that the most valid method of measuring peak lower-body mechanical power output (LBPP) in a countermovement jump (CMJ) is by analysis of the corresponding vertical component of the ground reaction force (VGRF)—time history of the jump. However, there is no published standard protocol. The purpose of this study was to establish a standard protocol. The variables necessary to define a valid and reliable CMJ method were: (a) vertical force range, (b) force sampling and integration frequency, (c) method of integration, (d) determination of body weight (BW), and (e) determination of the initiation of the CMJ. Countermovement jumps off a force platform (FP) were performed by 15 male professional rugby players. The 5 variables were then optimized to maximize the reliability and validity of the measure of LBPP. Errors of <1% ($p \leq 0.05$) in the measurement of LBPP were obtained using the following specification: (a) 6 times BW (using a 16-bit analog to digital converter), (b) 1,000 Hz, (c) Simpson's rule or the trapezoidal rule, (d) mean VGRF for 1 second of quiet standing immediately before jump signal, and (e) 30 ms before the instant $BW \pm 5 SD$ is exceeded after the jump signal. Peak lower-body power output was most sensitive to variables 4 and 5. It was concluded that this study has established a standard protocol for the criterion method of measuring peak power in a CMJ using an FP. As all other estimates and less reliable methods of determining LBPP in a CMJ rely on the FP method for calibration, it is proposed that this protocol be used as the basis of future criterion measures using a FP.

KEY WORDS muscular power, force platform, training interventions, reliability, component forces

INTRODUCTION

Muscular power is widely considered to be a key determinant in athletic performance, particularly in sports that require large amounts of force generation in a short period of time (6,13,22). As such, power development for athletes is an essential element of their training programs (2,12,17). Sports that require high acceleration, sprinting, or jumping require powerful lower-body movements; consequently, a reliable and valid measure of lower-body power is necessary to monitor the effect of training interventions and readiness to train. In the case of peak instantaneous lower-body mechanical power output (LBPP), the countermovement jump (CMJ) has been used for many decades and can be performed with a number of variations (4,15,17,19).

The criterion (or reference) method of measuring LBPP, used to validate estimates or as a “gold standard” measure, is based on performance in a vertical jump, where the subject is required to jump off a force platform (FP) (4,9,12,14). The vertical component of the ground reaction force (VGRF) is recorded from the time just before the start of the jump until after take-off. The impulse momentum relationship is applied, at the sample rate of the FP, to the net vertical force to determine instantaneous velocity of the whole body centre of gravity (CG). Instantaneous power (P) is then calculated from the product of the force (F) and velocity (v): $P = F \times v$ (7,21).

In order for the criterion method to achieve maximum validity and reliability, it is necessary to control and specify several key variables: vertical force range and resolution, force sampling frequency and resultant force integration frequency, method of integration, determination of body weight (BW), and determination of the initiation of the jump. Whereas some investigators have considered the effect of some of these key variables, for example, the effect of varying sampling frequency on several jump performance measurements including peak instantaneous power (11) and the effect of different jump start thresholds on kinematics and kinetics of a CMJ (16), there do not appear to be any published studies that have reported the effect of a combination of all key variables. Consequently, no criterion protocol

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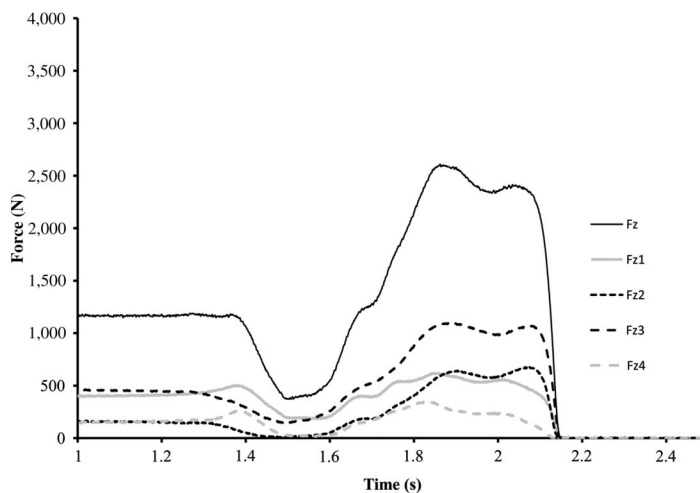


Figure 1. Vertical force-time history of a countermovement jump showing the 4 corner vertical force components. Fz = resultant vertical force Fz1–Fz4 are corner vertical force components or generally Fzc.

currently exists. Therefore, the purpose of this study was to establish a protocol for the measurement of LBPP produced during a CMJ using the criterion FP method. The protocol was established using the key variables of vertical force range (collected at 16-bit resolution), force sampling frequency and resultant force integration frequency, method of integration, determination of BW, and determination of the initiation of the jump.

METHODS

Experimental Approach to the Problem

Force-time histories were collected for 15 professional rugby players, each performing 1 CMJ with arms akimbo (hands on hips). The force-time histories were then used to determine the influence of vertical force range, sampling frequency, determination of BW, identification of jump initiation, and method of numerical integration on LBPP of each subject's CG by systematically varying each 1 and monitoring the effect on LBPP.

TABLE 1. Vertical forces produced during a countermovement jump.*

	Fz max (N)	Fzc max (N)	Body weight (N)
Minimum	2,060	770	799
Maximum	2,950	1,210	1,166
Mean	2,458	988	1,005
SD	260	145	121

*Fz max = maximum resultant vertical ground reaction force; Fzc max = maximum of the 4 corner component vertical forces.

Body mass (BM) was determined from BW and taken to be $BW \cdot g^{-1}$ (kg) with g = acceleration due to gravity.

Subjects

The participants were 15 male international rugby union players (age = 25.8 ± 3.4 years, stature = 1.89 ± 0.09 m, mass = 102.5 ± 13.9 kg) who were familiar with CMJs, as they formed an element of their training and testing regime. The University Ethics Committee approved all experimental procedures, and all participants were volunteers and gave informed written consent.

Experimental Procedure

The participants undertook the tests at the beginning of a pre-season training session. All participants underwent the same warm-up consisting of 5 minutes of light cycling followed by dynamic movements with emphasis placed on the musculature associated with jumping. After warming up, participants completed 1 maximal CMJ. All participants were given standardized instructions to stand on the FP (model number 92866AA, Kistler Instruments Ltd., Farnborough, United Kingdom) and jump when a signal lamp illuminated. The platform's vertical range was set to its maximum, 0–20 kN (i.e., 0–5 kN per corner transducer). The signal to jump and data collection were triggered when the participant remained stationary for more than 1 second, as observed by the operator. This ensuring that the first second of data collection (1 second of pretrigger) could be used to accurately determine BW, and as a reference to from which to determine a jump initiation time. The analog signal from the FP was sampled at a frequency of 1,000 Hz through a 16-bit analog to digital converter (ADC) using Kistler's Bioware (version 3.2.7.0; Kistler Instruments Ltd., Farnborough, United Kingdom). An initial sampling frequency of 1,000 Hz was chosen because it is the highest sampling frequency normally used to measure LBPP in a CMJ (2,8,14). A sample length of 5 seconds was used for all jumps, including the 1-second pretrigger phase. The FP was factory calibrated and before testing underwent satisfactory calibration checks using masses that were traceable to national standards.

Measurements

Calculation of Power. Power was determined from the unfiltered force-time history using the impulse momentum principle. The resultant VGRF was numerically integrated at the sample frequency of the force-time history and divided by BM to determine instantaneous velocity for time points

TABLE 2. Mean and SD of body weights determined at different sampling frequencies during 1 second of quiet standing ($n = 15$).

	Body weight (N)		
	1,000 Hz	500 Hz	100 Hz
Mean	1005.4	1005.4	1005.3
SD	136.7	136.7	136.8

that corresponded with the original force–time history. Instantaneous power was then taken as the product of the instantaneous velocity and VGRF at corresponding time points.

Analysis of Vertical Force Range and Body Weight. The VGRF measured by a FP consists of the arithmetic sum of 4 individual vertical force signals originating from the 4 transducers of the platform (1 transducer mounted in each corner) (Figure 1). Consequently, it was necessary to consider the force transmitted through each corner transducer as well as the combined, gross vertical force.

The vertical force–time histories for each participant's CMJ were recorded and the maximum unfiltered values of the gross force and the corner transducers' components of the gross force were determined by inspection for each subject. Body weight was taken to be the mean value of the VGRF during the first second of data collection. Body weights for the resampled data (500 and 100 Hz) were determined using the same procedure.

Analysis of the Use of Different Sampling Frequencies. To investigate the effect of sampling frequency on the determination of

LBPP from performance in a CMJ, force–time histories recorded for the vertical force range analysis were used. These 15 force–time histories were then resampled using Bioware's resampling function at 500 and 100 Hz. The LBPP was determined for each jump at the 3 unfiltered sampling frequencies using Simpson's rule at the corresponding frequency to determine the velocity–time data. Body weight was defined as mean VGRF during 1 second of stance. The integration start time, t_s , was defined as the point when the VGRF, after a signal to jump had been given, exceeded BW plus or minus 5 SD. The instant t_i was not optimized; however, it served as an initial reference start time. Consequently, the same method (incorporating the determination of BW, integration start time, and Simpson's rule) was used to determine LBPP for all jumps, differences in peak power for each jump could be attributed to the different sampling frequencies.

Analysis of Method of Numerical Integration. To investigate the effect of the method of integration on LBPP from a CMJ, the 15 unfiltered force–time histories sampled at 1,000 Hz were used. The start point for the integration was taken as t_s (defined above) and each force–time history was integrated twice, first using Simpson's rule and then using the trapezoidal rule, at the sampling frequency to determine the velocity–time data and hence mechanical vertical power output. The LBPPs for all jumps were determined by inspection for both methods of integration.

Analysis of Identification of the Initiation of a Countermovement Jump. During the stance phase of a CMJ, the VGRF will vary constantly because of slight movement of the subject and noise in the instrumentation, both internal and external. Therefore, it was necessary to define a threshold value of VGRF during the stance phase, beyond which the jump was defined as having been initiated. If the threshold was set too low, a mistriggering could occur, if set too high the defined initiation of the jump would be too late. Thus the initiation time, t_s , was defined as

the instant, after the signal to jump has been given, that the VGRF exceeded the mean plus or minus 5 SD of the BW as measured in the stance phase. This threshold would reduce the probability of mistriggering in the stance phase to $p < 0.0000006$ (i.e., 1 mistrigger every 1,744 jumps on average, for a stance phase of 1-second sampled at 1,000 Hz).

However, there is similarly a high probability that t_s , as described above, while identifying an instant that has a very low probability of being part of the stance phase, will be in

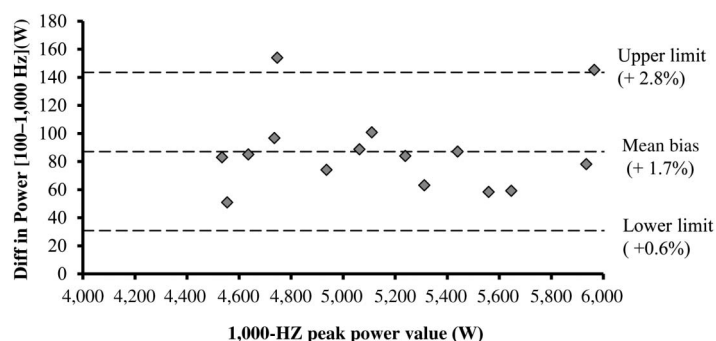


Figure 2. Bland and Altman plot comparing peak vertical mechanical power outputs of countermovement jumps using sampling frequencies of 100 and 1,000 Hz.

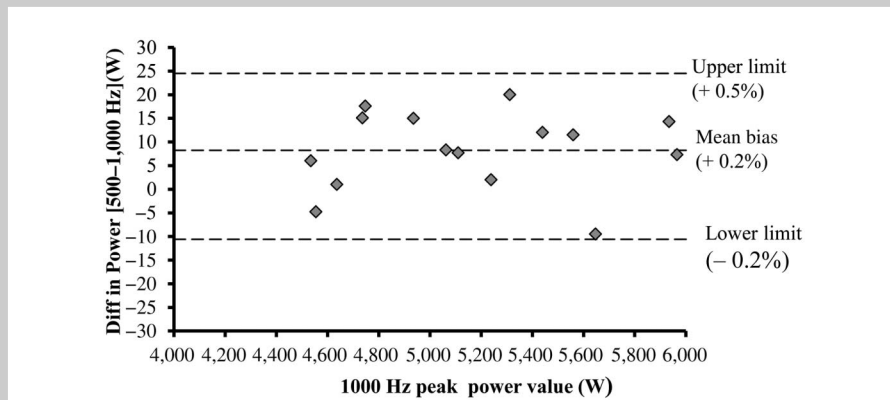


Figure 3. Bland and Altman plot comparing peak vertical mechanical power outputs of countermovement jumps using sampling frequencies of 500 and 1,000 Hz.

the jump phase of the CMJ. Therefore, to investigate the effect of varying t_s , and consequently its suitability as a start point, t_s was identified for the 15 CMJ's unfiltered force-time histories. The LBPP was determined using an integration starting point equal to, $t_s = 100$ ms, for each subject. The point, $t_s = 100$ ms was chosen because it was clearly in the stationary phase of the jump. Values of LBPP were then determined using integration starting points of $t_s = 90$ ms through t_s , to, $t_s + 30$ ms at intervals of 10 ms for each participant. Consequently any within-participant variation in LBPP could be attributed to the integration starting point.

Statistical Analyses

All statistical analysis was performed using SPSS 19 (SPSS Inc., Chicago, IL, USA).

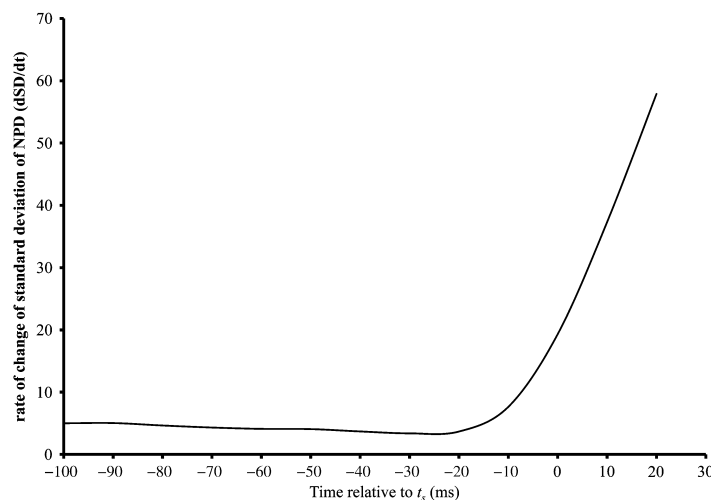


Figure 4. Graph of the rate of change of the SD of NPD.

Analysis of a Vertical Force Range and Body Weight. Mean and SD were determined for the vertical force-time histories' maximum values of the gross force and the corner transducers' components of the gross force. Mean and SD were determined for values of BW at sampling frequencies of 100, 500, and 1,000 Hz.

Analysis of the Use of Different Sampling Frequencies. To investigate the difference in LBPP for different sampling frequencies, limits of agreement and mean systematic bias of power output produced by the 100

and 500 Hz sampling frequencies, in relation to the power outputs of the 1,000 Hz sampling frequency were assessed using Bland and Altman plots (Bland and Altman (3)).

Analysis of Method of Numerical Integration. Limits of agreement and mean bias of LBPP produced by the 2 methods of numerical integration were assessed using Bland and Altman plots (3). It was unclear which of the 2 methods of integration produced the more accurate result that is, whereas the trapezoidal rule will exactly measure the area of the trapezoids produced by discrete sampling, Simpson's rule may produce a curve that better fits the analog VGRF time-history. Consequently the best estimate of actual peak power was taken as the mean of the 2 measurements (3). The difference values were determined by subtracting the trapezoidal rule values and the Simpson's rule values from the mean.

Analysis of Identification of the Initiation of a Countermovement Jump. The LBPP determined with an integration start point of $t_s = 100$ ms was taken as an LBPP reference value (PRV). For each subject, difference values were determined for LBPP by subtracting each LBPP determined using integration starting points of $t_s = 90$ ms through to $t_s + 30$ ms from PRV. The values were then normalized to PRV and expressed as a percentage and termed the normalized percentage difference (NPD). For each time point, $t_s = 90$ ms through to $t_s + 30$ ms, mean

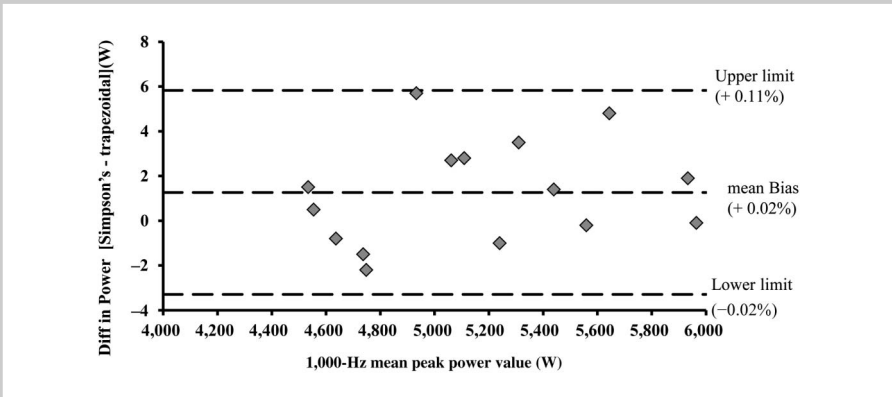


Figure 5. Bland and Altman plot comparing peak vertical mechanical power outputs of countermovement jumps using Simpson's rule and the trapezoidal rule.

values and *SD* were determined for the NPDs. Thus, the mean ± 3 *SD* was taken to represent the difference value of LBPP, plus associated uncertainty, for any integration start point after $t_s = 100$ ms ($p \leq 0.003$) as compared with PRV. To investigate the rate of change in the uncertainty of the difference value of LBPP, the first derivative of the *SD* of time series NPD (dSD/dt) was numerically determined giving the rate of change of uncertainty in the NPD. If an integration start point was taken after the jump had started, there would clearly be an increase in the NPD because the integration process would not have included part of the jump. Equally, as the integration start point moves further into the jump, the rate of change of uncertainty would increase rapidly. Hence, it is reasonable to expect the first derivative of the time series of *SD* of NPD to identify when the jump had started. It was necessary to analyze the *SD* of the NPD because the mean value NPD has the potential to show very little change at the beginning of a CMJ, as approximately half of the participants executing a CMJ start by first

moving up that is, a counter-movement before the CMJ, and half start by immediately dipping. Thus at the start of a CMJ, it is likely that an increase in positive lower-body power would be mirrored by a corresponding increase in negative lower-body power in approximately half of participants. However dSD/dt is not sign-dependent and hence will identify a positive or a negative change in NPD.

RESULTS

Selection of Vertical Force Range

The maximum and minimum total vertical force, maximum and minimum vertical component forces, and mean and *SD* of all CMJs are displayed in Table 1. There was no difference in BW for the 3 different sample frequencies (to 1 d.p. ± 1 digit) (Table 2).

Selection of Sampling Frequency

The results of the comparison of sampling frequencies can be seen in Figures 2 and 3. The sampling frequency of 100 Hz, when compared with 1,000 Hz, produced a mean difference of 87 W and limits of agreement (mean $\pm 1.96 \times SD$) of 144 and 31 W. The sampling frequency of 500 Hz, when compared with 1,000 Hz, produced a mean difference of 8 W with limits of agreement of 24 and -11 W.

Method of Numerical Integration

Figure 5 shows the results of the integration methods comparison, the differences between LBPPs calculated using Simpson's rule, and the trapezoidal rule are plotted on the *y* axis, and the mean (mean between Simpson's rule value

TABLE 3. Criterion method specification for the measurement of peak mechanical power in a CMJ by the criterion force platform method.*†

Variable	Criterion method specification
Vertical force range and resolution	5.6 $\times$ BW or higher at 16-bit resolution
Sample frequency	1,000 Hz
Integration frequency	1,000 Hz
Method of integration	Simpson's rule or trapezoidal rule
Determination of body weight	Mean ground reaction force measured for 1 second of the stationary stance phase immediately before the signal to jump
Determination of initiation of jump	(The instant that BW ± 5 SD is exceeded after the signal to jump has been given) minus 30 ms

*CMJ = countermovement jump.
†A specification of BW ± 5 SD as opposed to a reduction in BW for jump initiation was necessary because generally approximately half of all jumpers start a CMJ by first raising their centre of gravity.

and trapezoidal rule value) is plotted on the x axis. The analysis resulted in a mean of the difference of 13 W and limits of agreement (mean $\pm 1.96 \times SD$) of 6 and 19 W.

Identification of the Initiation of a Countermovement Jump

Figure 4 shows the rate of change of NPD. The graph shows a slight negative gradient between $t_s = 90$ to $t_s = 30$. At a point between $t_s = 30$ and $t_s = 20$, an inflection point occurs, thereafter the gradient increases rapidly.

DISCUSSION

Body weight was determined by taking the mean GRF value, as measured by the FP, for 1 second of the stance phase immediately before the signal to jump being given. In this study, sampling frequency (100, 500, and 1,000 Hz) had no effect on the determination of BW (to 1 d.p. ± 1 digit) (Table 2).

The maximum resultant VGRF recorded in all trials was 2,950 N in a jump by a subject with a BW of 1,166 N (Table 1). This is consistent with Kibele (14) who reported that maximum vertical VGRF during a CMJ were in the region of 3–3.5 times BW. However, Kibele (14) did not report component vertical loads. Failure to considering component loads can lead to errors because of the range of individual force transducers being exceeded. If, for example, the total vertical force range in this study had been set on the basis of 3.5 times the BW of the highest weight subject (1,166 N), this would give a maximum expected vertical force of 4,081 N, that is 3.5 times 1,166 N, corresponding to a maximum range of 1,020 N (4,081 N/4) for each component force transducer. This value would have been exceeded in 1 or more component force transducers in 47% of the jumps causing an erroneous force reading. An error of this sort would not be obvious from the resultant vertical force record because an overloaded component sensor would either produce a seemingly correct force–time history but out of the calibrated range or if the absolute maximum of the transducer had been reached, a plateaued force–time history. In both cases, when the component transducers' outputs had been summed, the error would not be apparent. A more robust method of specifying the maximum force range is to determine the maximum value for the component transducers. This can be calculated empirically from the study data. The range for this study was defined as the mean maximum vertical component force plus 3 SD , $988 \text{ N} + (145 \text{ N} \times 3) = 1,423 \text{ N}$. The corresponding resultant maximum vertical force range for the FP would then be $1,423 \text{ N} \times 4 = 5,692 \text{ N}$, that is 5.7 times BW. Setting a FP's range to this value, or higher, would reduce the probability of it being exceeded to $p \leq 0.003$.

Generally, when sampling a signal to represent it elsewhere, the higher the sampling frequency the greater the fidelity of the representation of the original signal. Specifically Nyquist's sampling theorem (18) states that a sampling frequency of double the highest frequency contained in the signal is necessary to ensure that none of the original signal is lost during the sampling process and also to prevent aliasing. The signal

of interest in this study was the force–time history of a CMJ. Usually, Fourier's analysis is used to determine the highest frequency present in a signal, however a force–time history cannot be represented by a function and is noncyclical and as such is not suitable for this type of analysis. On this basis, an arbitrary initial sampling rate of 1,000 Hz was chosen for this study. It can be reasonably assumed that the mean difference and limits of agreement between a 1,000 Hz sampling frequency and a 2,000 Hz sampling frequency would be at least as good as, or better than, those obtained for the comparison between 500 and 1,000 Hz. This being the case, there would be no need to sample at 2,000 Hz because a sampling frequency of 1,000 Hz would achieve precision of $<1\%$ (Figure 3). It is also highly likely that 500 Hz would also achieve this precision. The worst case scenario for the precision of a sampling frequency of 500 Hz would be that the mean difference and limits of agreement of 1,000 Hz sampling frequency compared with 2,000 Hz sampling frequency were the same as for 500 Hz compared with 1,000 Hz giving a mean difference between 500 and 2,000 Hz of $+0.2\%$, with an upper limit of agreement of $+1.0\%$ and a lower limit of agreement of -0.4% . However, as a sampling frequency of 1,000 Hz produces more accurate results than a sampling frequency of 500 Hz, and given the convenience of sampling in time intervals of milliseconds, a sampling frequency of 1,000 Hz was chosen as the preferred frequency for the determination of power output by the criterion FP method in this study.

There appears to be no agreed method in the literature of determining the instant when a CMJ has been initiated. Some researchers use a relative threshold such as 5% BW during the stance phase and define the jump initiation as the instant that the VGRF falls below that threshold (5,11,20). Others qualitatively assess where the jump has started by manually inspecting the force trace (8) or refer to software determination but do not describe methods (14,9). Some researchers do not report any methods (1,10). None of these methods are ideal, although the first is reliable, it clearly does not retain the entire jump signal, others are either subjective or not defined or not reported. The method described here sought to minimize the uncertainty in peak power by identifying an instant, such that the entire jump signal was retained but none of the stance phase. The rate of change of the SD of the NPD shows a slight negative gradient between times $t_s = 90$ ms and $t_s = 30$ ms (Figure 4), indicating that there is a small constant variation in the SD of the NPD corresponding to instrument drift and small unbalanced impulses. Between $t_s = 30$ and $t_s = 20$ ms, an inflection point occurs and the rate of change of SD of the NPD starts to increase rapidly, indicating that starting integration after this point would miss some of the beginning of the CMJ. Therefore, the initiation of the CMJ must be before $t_s = 20$ ms and hence $t_s = 30$ ms can be identified as a valid initiation point of a CMJ.

Figure 5 shows that the maximum error, ΔP , in the determination of peak power between Simpson's rule and the trapezoidal rule would be, $\Delta P \leq 0.13\%$ (confidence interval = 95%).

However, because it is unclear which of these methods of integration gives the more correct result, the best estimate of the correct value of peak power can be taken as the mean of the 2 results. It can therefore be concluded that if a maximum error of 0.13% in the determination of peak power is acceptable, then the 2 methods of numerical integration, Simpson's rule, and the trapezoidal rule can be used interchangeably. Based on the above conclusions, Table 3, the criterion method for determining peak vertical mechanical power output in a CMJ was defined.

This study did not investigate the effect of ADC resolution on LBPP because generally it is not adjustable in commercial systems. However, it will clearly affect the determination of LBPP. The 2 most common resolutions currently in use are 12 bit (e.g., AMTI) and 16 bit (e.g., Kistler and Bertec). A 12-bit ADC is capable of representing an analog signal as 4,096 that is 2^{12} , discrete steps, thus theoretically representing an analog force signal with a range of 0–20 kN in discrete steps of 4.9 N, whereas a 16-bit ADC would theoretically represent the same signal with a resolution of 0.3 N. However, in practice, the resolution of a system is dependent on other factors in addition to ADC bits including system noise and actual force range as opposed to stated maximum range. For example, the Kistler 9281E FP has a maximum VGRF range of 20 kN, however the absolute range of the platform is 30 kN because it is also capable of measuring a vertical "pull" of 10 kN. Clearly, this would not be the case with a portable system. It is reasonable to expect a 16-bit ADC to better represent an analog signal than a 12-bit ADC, therefore, whenever possible, a 16-bit ADC should be used in preference to 12-bit ADC, particularly when high force ranges are used. The absolute range of the platform should also be clearly stated.

PRACTICAL APPLICATIONS

Testing peak mechanical power in a CMJ using a FP is widely used as a method of monitoring and assessing elite athletes. The criterion method described in this study, Table 3, is valid and reliable and could be adopted as the basis of a standard method of testing LBPP in a CMJ when using a FP.

FURTHER RESEARCH

Peak vertical mechanical power output in a CMJ as measured by a FP is one of a number of performance variables (probably the most common) that are used in the assessment of elite athletes. Future work should focus on the influence of ADC resolution on LBPP and also developing criterion methods for the determination of other commonly used variables derived from force–time histories of a CMJ, for example, rate of force development (both eccentric and concentric), velocity, and jump spring stiffness.

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